APPLIED ECOLOGY:
NEW TOOLS & DATA STREAMS

GROUP 10

CITIZEN SCIENCE

- Citizen Science: Public participation in scientific data collection and problem-solving.

- 1600s Origins: Amateur explorers built early scientific foundations.

1800s Shift: Science professionalized; universities excluded the public from research.

CITIZEN SCIENCE

- The Reality: Nature's scale exceeds the reach of isolated university labs.

- 1900 Milestone: Launch of the Audubon Christmas Bird Count.

CHRISTMAS BIRD CENSUS

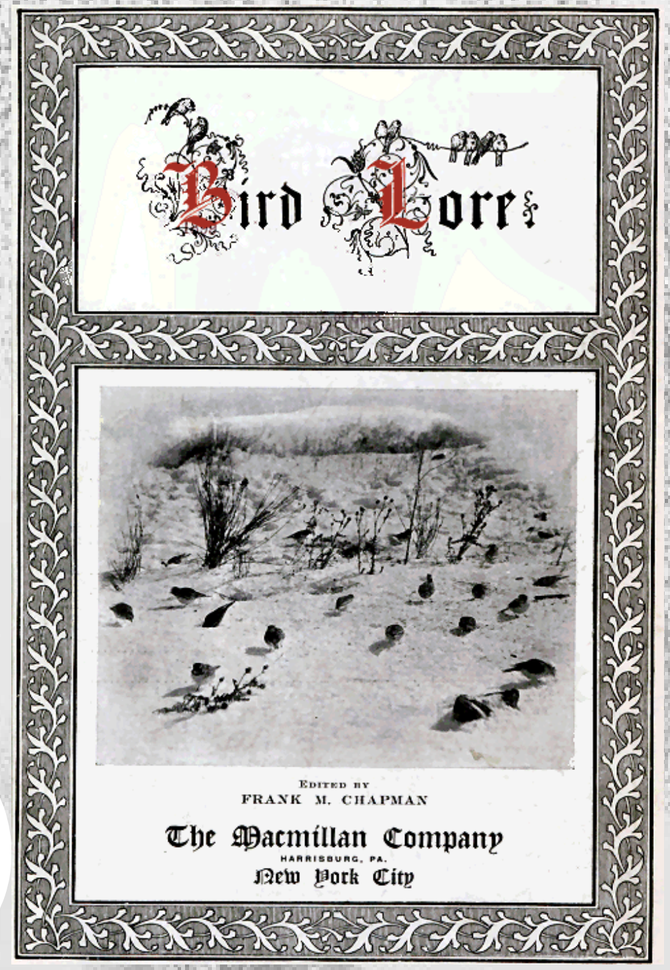
- Provides data about populations of birds for more than 125 years.

- Volunteers from North America count every bird they see or hear in a designated area within a 24-hour period.

https://www.audubon.org/magazine/lifer-warblers-highland-rarities-guatemala-christmas-bird-count-was-full-of-thrills

CHRISTMAS BIRD CENSUS

- *CBC was first proposed in 1900 by US ornitologist Frank Chapman*



The Christmas Bird Census 29

It has not been possible to publish all the notes which have been received, and it has been necessary to omit the descriptive matter which, in some instances, was given. At another time we hope to suggest a bird census on somewhat more exact lines, through the enumeration, not of the birds observed on a given day, but of those occupying a certain area, when a description of the character of the ground, etc., will be of importance.

SCOTCH LAKE, YORK COUNTY, NEW BRUNSWICK

Time, 9 A. M. to 10 A. M. Clear; wind, west, light; temp., 32°.

Goshawk, 1; Hairy Woodpecker, 1; Downy Woodpecker, 1; Blue Jay, 2; Pine Grosbeak, 1; Brown Creeper, 2; White-breasted Nuthatch, 20; Chickadee, 6. Total, 9 species, 36 individuals.—WM. H. MOORE.

TORONTO, CANADA

Time, 11.30 A. M. to 4.30 P. M. Clear; wind, northwest, 10 miles per hour; temp., 25°.

Crow, 15; White-breasted Nuthatch, 1; Chickadee, 10; Golden-Crowned Kinglet, 25. Total, 4 species, 41 individuals.—E. FANNIE JONES.

KEENE, N. H.

Time, 10.30 A. M. to 1.30 P. M. Clear; wind, northwest, very light; temp., 32°.

Crow, 1; Northern Shrike, 1; Chickadee, 16. Total, 3 species, 18 individuals.—CLARENCE MORRISON BROOKS.

BELMONT AND CAMBRIDGE, MASS.

Time, 7.30 A. M. to 11.30 A. M. Clear; wind, northwest, very light; temp., 34°.

Fresh Pond, Cambridge.—American Herring Gull, according to W. Deane, there were 250 on the pond at 9.30 A. M. Flocks kept coming in all the morning until, at 11.30, there were 1,375; Black-backed Gull, 2; Black Duck, 90; Mallard, 1; Tree Sparrow, 30; Song Sparrow, 14.

Belmont.—Flicker, 7; Downy Woodpecker, 2; Blue Jay, 10; Crow, 17; American Crossbill, 1; Purple Finch, 2; Junco, 10; Northern Shrike, 1; White-breasted Nuthatch, 4; Golden-crowned Kinglet, 10; Chickadee, 17. Total, 17 species, 1,593 individuals.—RALPH HOFFMANN.

BOSTON, MASS. (ARNOLD ARBORETUM)

December 26, 9.30 to 2.30. Clear, snow-squall, clear; wind, southwest to northwest, light; temp., 37°.

Bob White, 12–15; Flicker, 6, 7; Blue Jay, 15–18; Crow, 16; American Crossbill, 1; Goldfinch, 18; Tree Sparrow, 5, 6; Song Sparrow, 2; Junco, 20–25; Myrtle Warbler, 7; Brown Creeper, 2; White-breasted Nuthatch, 2; Chickadee, 12–15. Total, 13 species, about 125 individuals.

On December 19, in these grounds, there were also Northern Shrike, 1; Golden-crowned Kinglet, 2; White-throated Sparrow, 2; Fox Sparrow, 1.—HORACE W. WRIGHT.

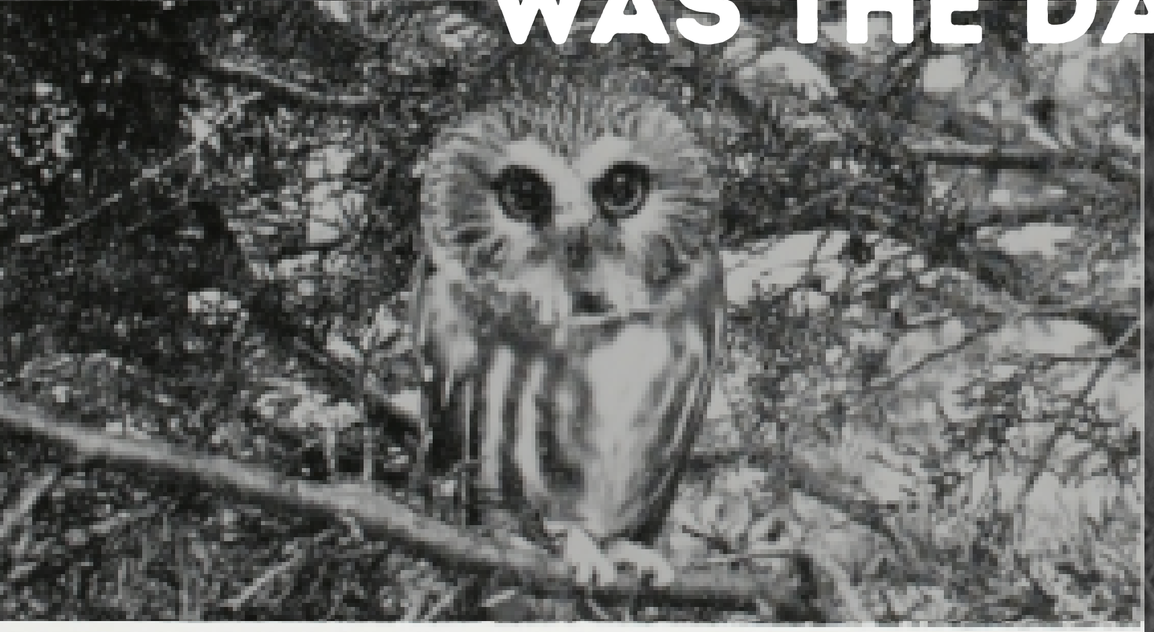
Digitized by Google
Original from UNIVERSITY OF COLORADO BOULDER

https://www.britannica.com/biography/Frank-M-Chapman

https://www.gutenberg.org/files/47500/47500-h/47500-h.htm

https://www.gutenberg.org/files/47500/47500-h/47500-h.htm

WAS THE DATA RELIABLE?



IT'S A RED-LETTER DAY WHEN THE SAW-WHET OWL IS FOUND ON A BIRD CENSUS TRIP
(Fred W. Kent photograph)

SPIPOLL

- A project studying insects and pollination.
- Volunteers photograph all insect visitors on a chosen plant for 20 minutes.
- Photos are uploaded and identified using an online guide.

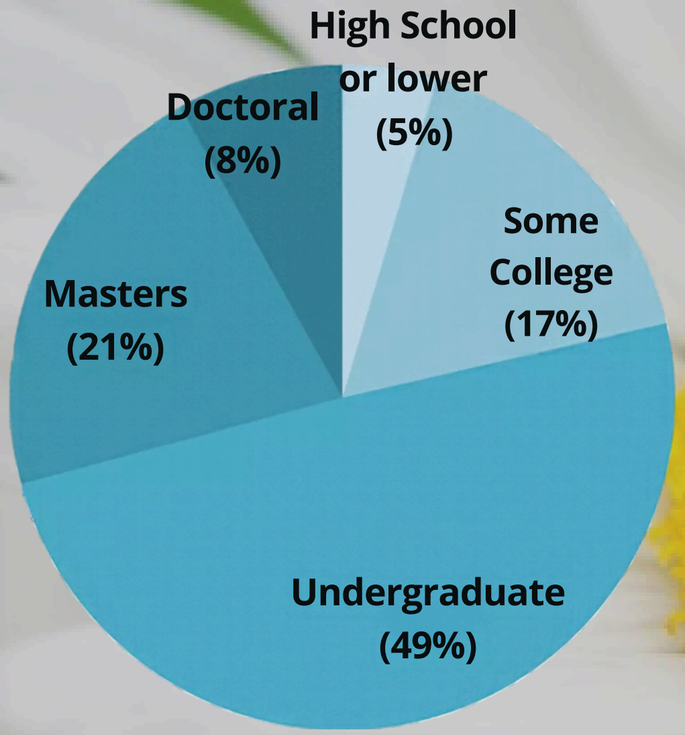
- 5627 participants
- 97605 collections

citizen science was still not formally recognized

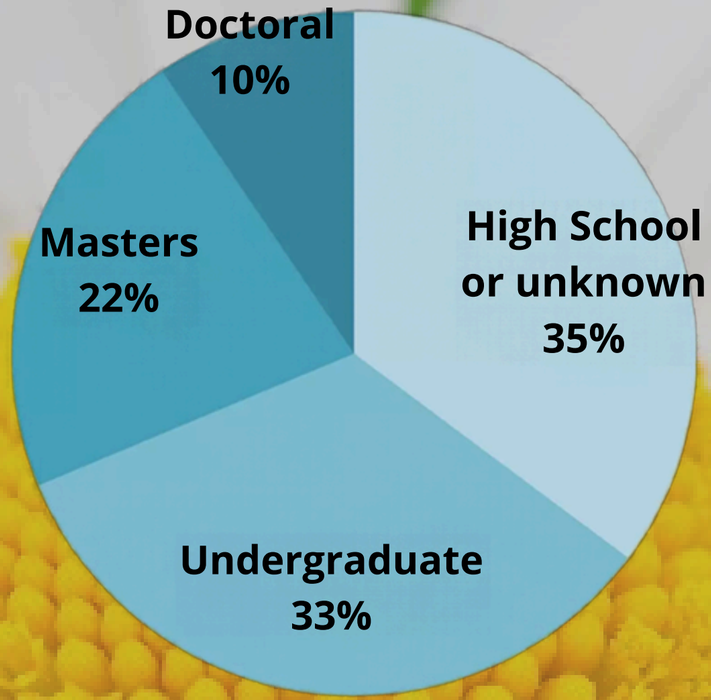
- underrepresentation in formal litterature
- not enough recognition of citizens contribution

Muki Haklay, proffesor of geography at UCL

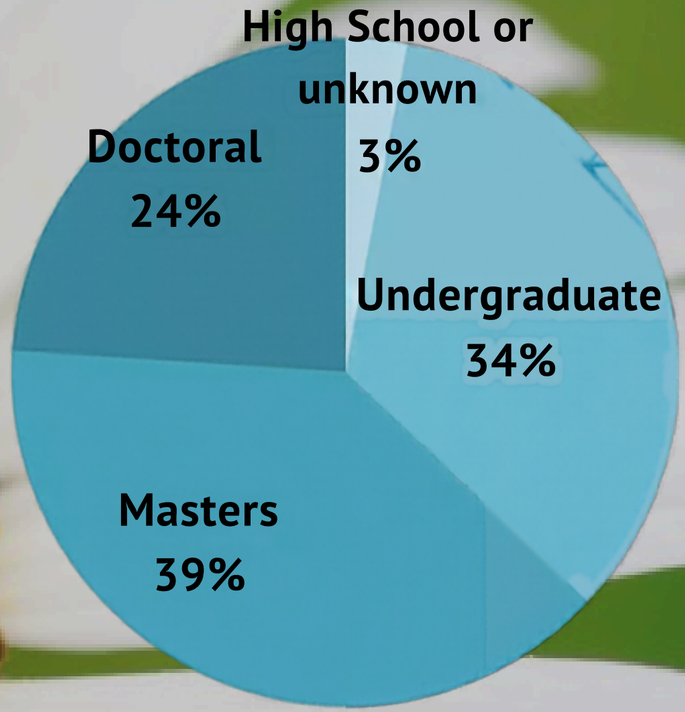
GRAPHS SHOWING EDUCATION OF PARTICIPANTS IN CITIZEN SCIENCE PROJECTS



OPEN STREET MAP



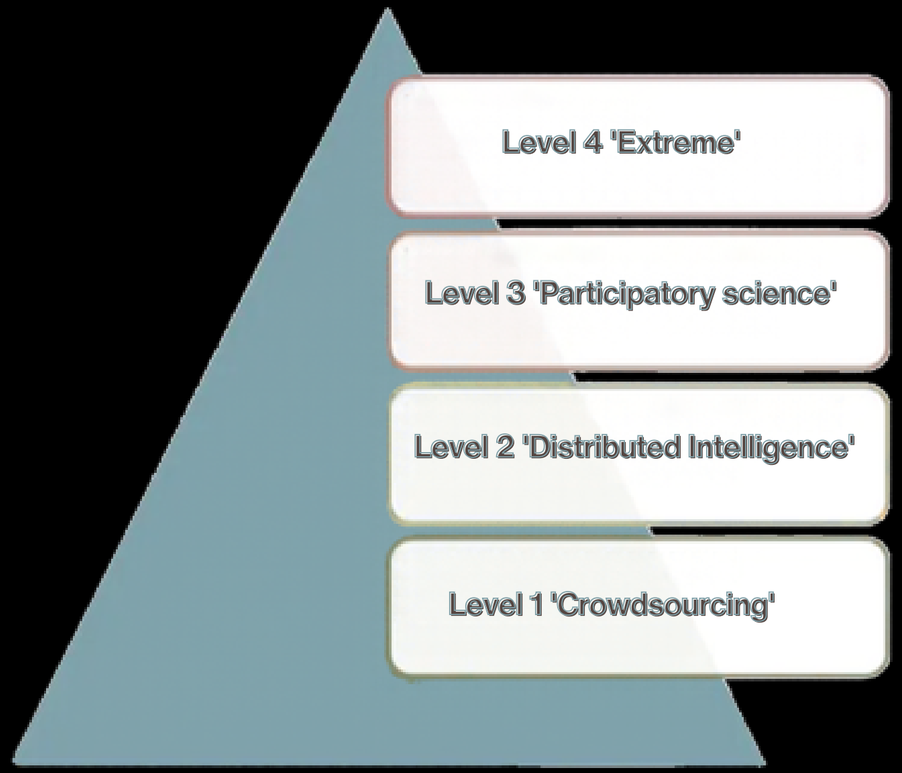
GALAXY ZOO



TRANSCRIBE BENTHAM

TYPOLOGY OF PARTICIATION

Levels of Citizen Science



Levels of Citizen Science 2011

Level 4 — Extreme Citizen Science
- Collaborative science - problem definition, data collection and analysis

Level 3 — Participatory Science
- Participation in problem definition and data collection

Level 2 — Distributed Intelligence
- Citizens as basic interpreters
- Volunteered thinking

Level 1 — Crowdsourcing
- Citizens as sensors
- Volunteered computing

Citizen Science and Volunteered Geographic Information – overview and typology of participation M. Haklay (2013)

Extreme Citizen Science (ExCiteS)

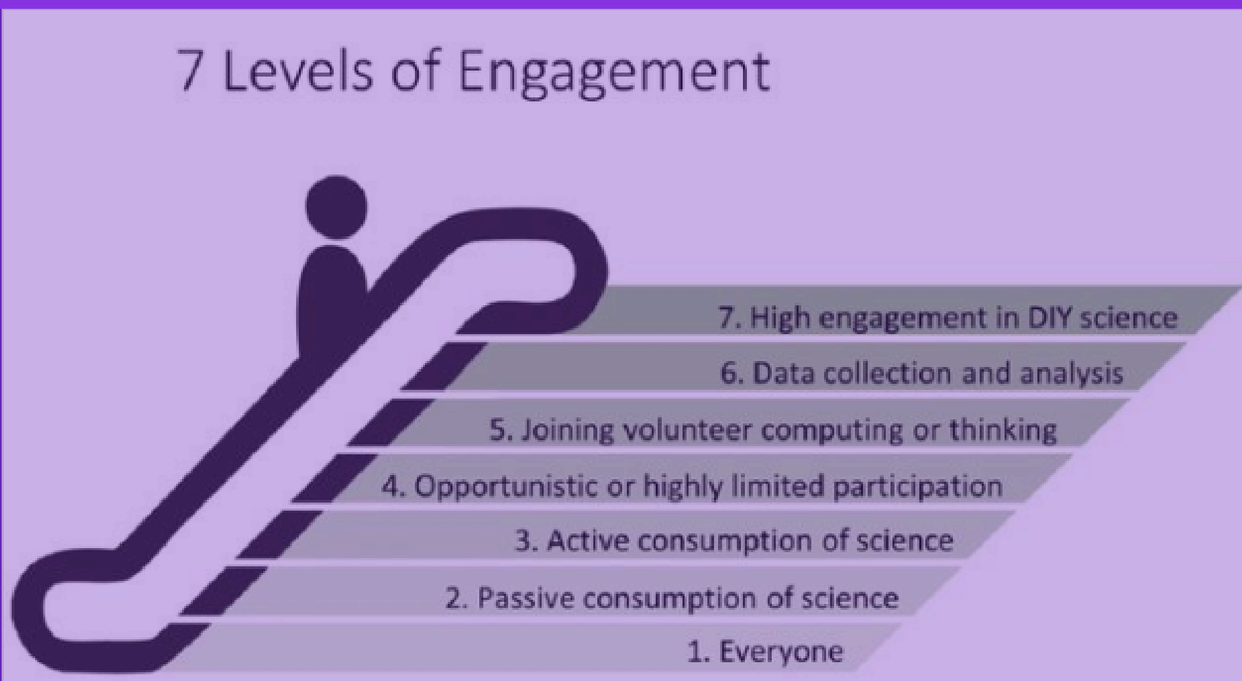


Figure 1: DITOs escalator framework for public engagement.

Download scientific diagram | DITOs escalator framework for public engagement.

10 PRINCIPLES
OF
CITIZEN SCIENCE

- Rapid expansion in absence of guidance during early 2010's
- By: european citizen science association (ECSA)
- Development : 2013 - 2015
- To: Set good practice
  Provide a framework
  Build credibility

ecsa | European Citizen Science Association

10 PRINCIPLES
OF
CITIZEN SCIENCE

1. ACTIVE CITIZEN INVOLVEMENT

- Public involvement in real scientific research
- Roles range from data contributors to project leaders

2.GENUINE SCIENCE OUTCOME

- Produces genuine scientific outcomes
- Informs research, conservation, and policy decisions

10 PRINCIPLES
OF
CITIZEN SCIENCE

3.MUTUAL BENEFITS

- Benefits both professional scientists and public participants
- Drives personal growth, scientific publication, and policy change

4.PARTICIPATION IN MUTLIPLE STAGES

- Opportunity to engage in multiple research stages
- Includes project design, data collection, analysis, and communication

10 PRINCIPLES
OF
CITIZEN SCIENCE

5.PARTICIPANT FEEDBACK

- Citizen scientists receive feedback from the project.
- For example, how their data are being used and what the research, policy or societal outcomes are.

6.ACKNOWLEDGMENT OF LIMITATIONS & BIASES

- A valid research method requiring standard bias control
- Uniquely boosts public engagement and democratizes science

10 PRINCIPLES
OF
CITIZEN SCIENCE

7.OPEN DATA & METADATA

- Data and results are published in open-access formats
- Shared publicly by default, barring privacy or security concerns

8. PROPER ACKNOWLEDGMENT

- Citizen scientists are acknowledged in project results and publications.

10 PRINCIPLES
OF
CITIZEN SCIENCE

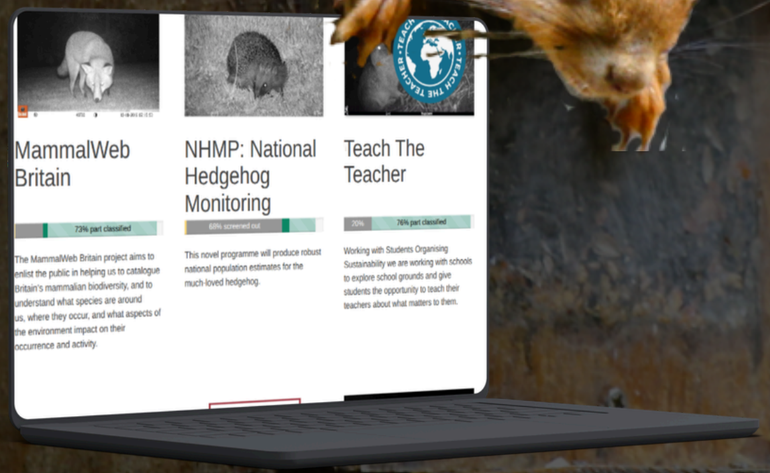
9.PROGRAMMATIC EVALUATION

- programmes are evaluated for their scientific output, data quality, participant experience and wider societal or policy impact

10.LEGAL & ETHICAL CONSIDERATION

- Prioritizes legal, ethical, and environmental responsibility
- Manages intellectual property, privacy, and data sharing

MAMMAL WEB

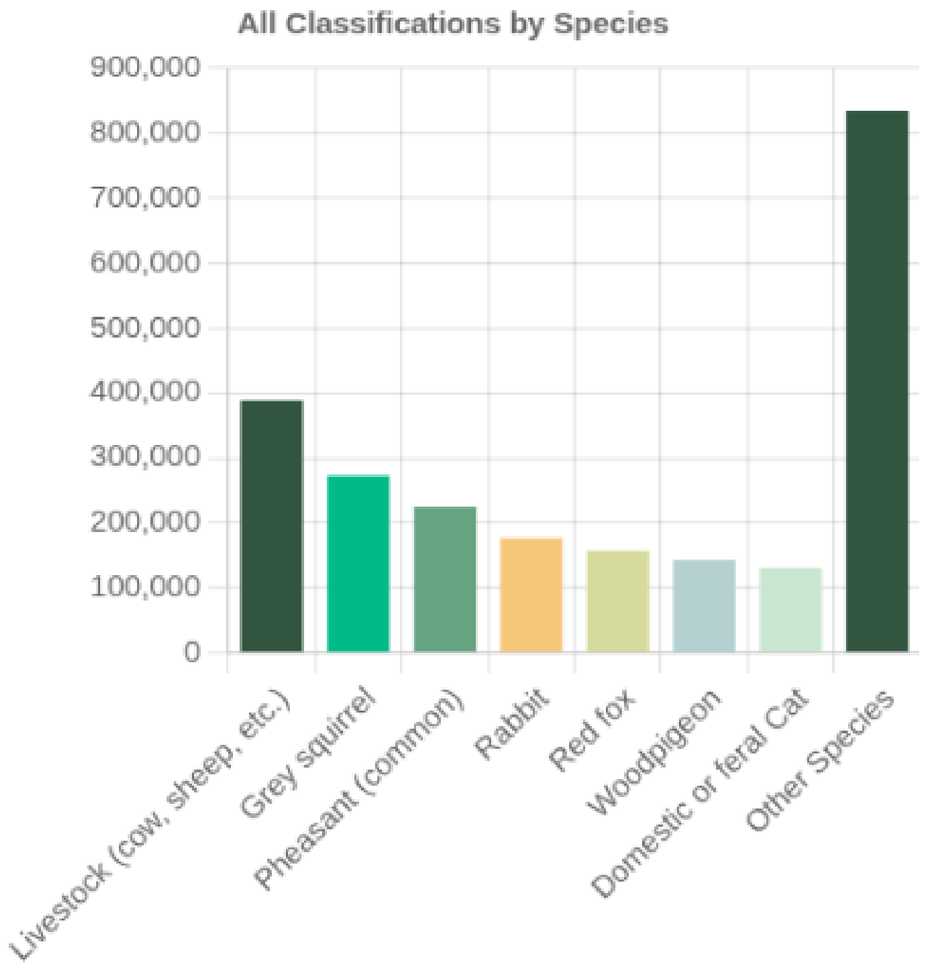


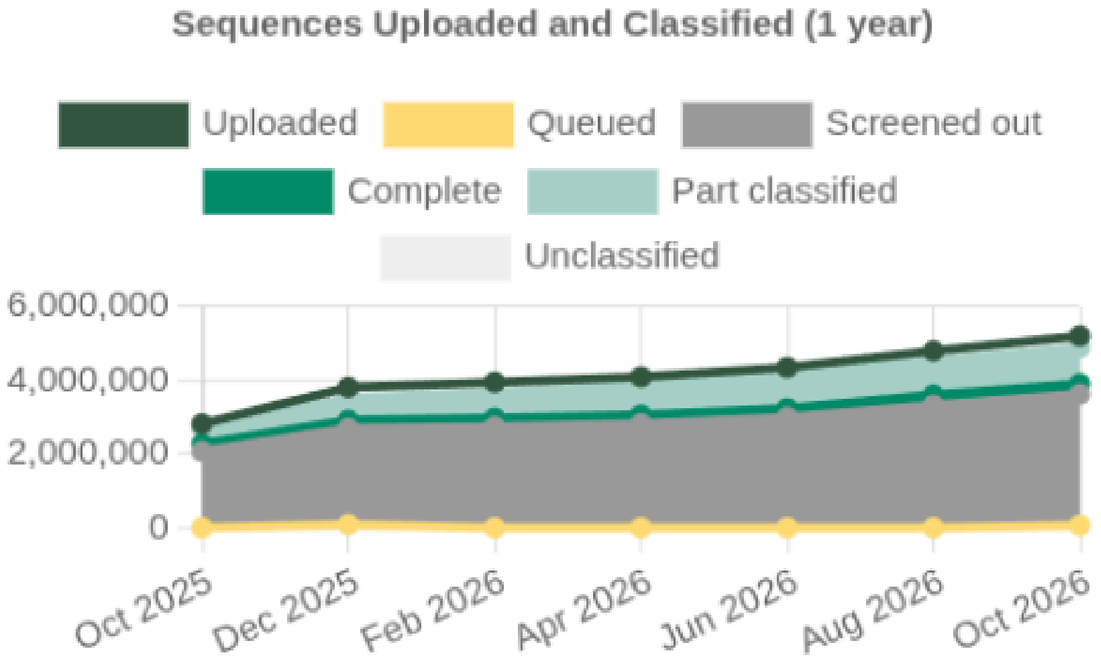
- Camera-trapping
- Centralized upload
- Privacy scrubbing
- Crowdsourced tagging
- Consensus verification
- Machine-learning automation
- Co-creation transition
- Nature reserve designation

https://www.magnific.com/free-photos-vectors/birds#uuid=422ea225-d0a7-469e-ba3b-b2426951dfe3

MAMMAL WEB

NHMP: NATIONAL HEDGEHOG MONITORING

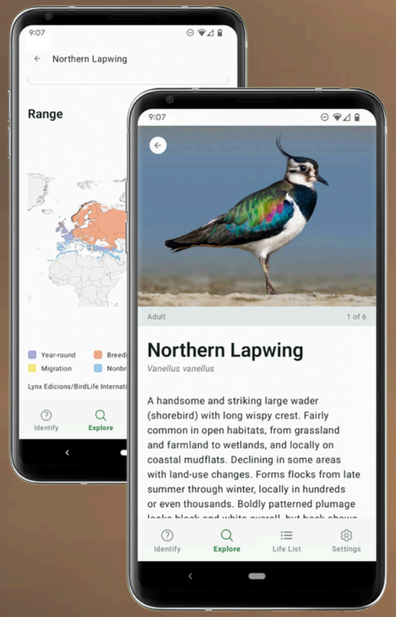




MOBILE TOOLS

Merlin Bird ID

by Cornell Lab

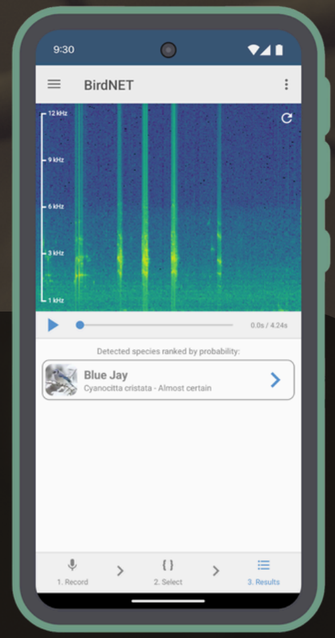


- Free, AI-powered app by the Cornell Lab of Ornithology.
- Sound ID: Real-time, offline bird song recognition.
- Photo ID: Instant species matches from your camera or gallery.

MOBILE TOOLS

BirdNET

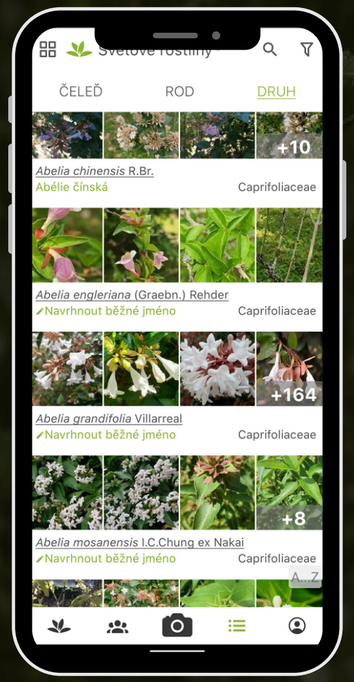
- Open-source AI recognizing 6,000+ bird species by sound.
- Creators: Cornell Lab of Ornithology & Chemnitz University.



- Capture: Splits audio into 3-second clips.
- Visualize: Converts sound into visual spectrograms.
- Analyze: Neural networks identify the species.
- Verify: Cross-references with location and date for accuracy

MOBILE TOOLS

PlantNet

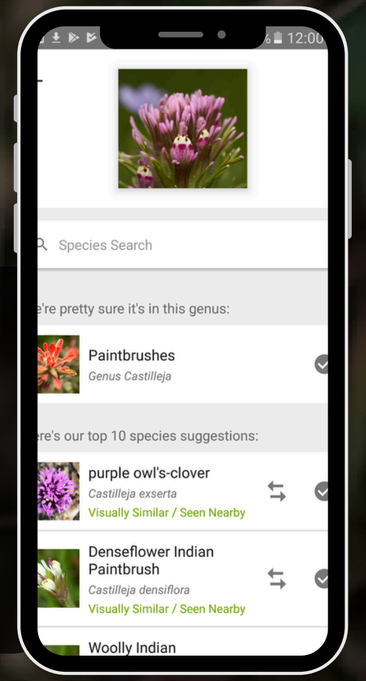


- Free app for automatic plant identification
- Recognizes ~20,000 species, prioritizing wild flora
- User photos help scientists track global biodiversity

MOBILE TOOLS

iNaturalist

- App for identifying plants, animals, and fungi using AI
- Findings verified by a community of 400k+ experts
- GPS-tagged media builds a global biodiversity database



CONCLUSION

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